

Deep Learning Assignment-1

Deep Learning (DL) is a specialized branch of artificial intelligence inspired by how the human brain processes information.

Instead of requiring humans to write rigid, step-by-step code telling a computer what to look for, DL relies on layered structures called **artificial neural networks**. You feed the system vast amounts of raw data like millions of photos, audio recordings, or text files and the network automatically learns to recognize patterns on its own.

The top 10 real world applications of DL are:

1. **Medical Imaging & Diagnostics:** Automated detection of tumours, fractures, and lesions in X-rays, CT scans, and MRIs, helping radiologists catch diseases earlier.
2. **Autonomous Vehicles:** Real-time object detection, lane tracking, pedestrian identification, and path planning in self-driving cars.
3. **Natural Language Processing (NLP):** Machine translation, automated document summarization, and sentiment analysis for customer feedback.
4. **Fraud Detection & Financial Security:** Continuous monitoring of transaction feeds to flag irregular credit card usage, account takeovers, and identity fraud.
5. **Recommendation Engines:** Personalized movie, product, and music suggestions on platforms like Netflix, Spotify, and Amazon based on user behaviour.
6. **Speech Recognition & Conversational AI:** Powering voice typing, real-time meeting transcription, and virtual assistants like Siri or Google Assistant.
7. **Biometrics & Facial Recognition:** Unlocking devices, identity verification at airports, and security surveillance.
8. **Drug Discovery & Genomics:** Predicting protein folding structures and simulating molecular interactions to speed up pharmaceutical research.
9. **Precision Agriculture:** Analysing satellite and drone imagery to spot crop diseases, monitor soil health, and predict harvest yields.
10. **Industrial Predictive Maintenance:** Analysing continuous sensor data (vibration, heat, pressure) from machinery to predict equipment failure before a breakdown occurs.

RECOMMENDATION ENGINE

In the early days of the web, platforms suggested content using basic filtering, if you bought a book, you were shown other books in the same category. Modern digital platforms like Netflix, Spotify, YouTube, and Amazon have moved far beyond these simple rules. Today, they rely on Deep Learning (DL) to power recommendation engines capable of parsing billions of items and personalizing feeds in real time for hundreds of millions of users.

Earlier recommendation systems relied primarily on *collaborative filtering* (matching users with similar watch histories) or *content-based filtering* (matching item attributes like genre or actor). While useful, these methods suffered from key limitations:

- **The "Cold Start" Problem:** Inability to recommend new items or serve new users with little historical data.
- **Sparsity:** Users interact with only a tiny fraction of a platform's total catalogue, leaving massive gaps in data tables.
- **Linearity:** Older statistical models could not capture complex, non-linear relationships.

Deep neural networks solve these challenges by projecting users, items, and contextual data into a shared mathematical space using embeddings (dense vector representations).

Instead of treating a movie simply as "Sci-Fi / 120 mins," a deep learning model learns hundreds of hidden mathematical features about that movie based on viewer habits. Similarly, the network converts a user's complex behaviour like watch duration, pause frequency, skips, time of day into a high-dimensional user profile.

To deliver recommendations in milliseconds across millions of choices, modern DL recommendation engines use a two-stage approach:

- **Candidate Generation** (Retrieval): The system takes a catalogue of millions of items and uses deep neural networks to quickly narrow them down to a few hundred relevant candidates. Models like Two-Tower Neural Networks treat user history as one input branch and item features as another, scoring broad compatibility efficiently.
- **Ranking:** The shortlisted candidates pass through a heavier, more complex deep network. This stage factors in fine-grained details such as network speed, current time, exact watch order, and item freshness to assign a precise probability score predicting whether the user will click, watch, or purchase.

Ultimately, DL turns massive catalogues into personalized feeds. Its ability to predict user intent makes recommendation engines a defining application of modern AI.